BIOMETRIC ATTENDANCE

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Abstract—**This paper is presented about a system of recording student attendance using fingerprint identification that allows students to monitor student attendance to class is a true electronically. It can reduce the presence of fraudulent students who are now mostly done by the students and the system can also reduce problems such as the presence of the missing paper and easily damaged. With this system can replace the existing manual system to a more systematic and electronics. This attendance system will be displayed on a computer lecturer with more attractive and graphics and have students complete detail using Microsoft Visual Basic Studio and integrated using the Fingerprint Reader.**

Index Terms—biometric, fingerprint, xbee

I. INTRODUCTION

Now, there are many universities around the country and each university consists of up to 10 thousand students. To handle a large number of students can be a problem especially for attendance. Now the processes to secure the attendance of the majority of universities are still using manual processes. Manual process means that when starting the class / lecture, the lecturer will give a piece of paper and students will check the presence of their names and then be signed. At the end of the class, the lecturer will take back the presence of paper and keep it as a record.

Nowadays, most of each university lecturer will provide a list of students to sign in for students who attend the class. Cheating often occurs in the presence of students. Therefore, if there is fraud student attendance, lecturer will not be able to identify the problem. For example, other students signed their other friends. So to avoid this problem, it is ideal for developing a attendance management system using biometric attendance will monitor and record the presence of each student in the class.

Typically, the presence of paper requires a lot of time for signing by all students, especially for classes with many students. Students forget to sign it and they assume the presence of class absence. This problem will also occur when the lecturer forget to bring paper to class attendance. Students should write their name on a piece of paper and sometimes the students will take a change for the presence of cheating in the process. Appropriate solution to this problem is to design a system that will record attendance automatically.

In this project, biometric system used to record student attendance automatically. This project will used student fingerprint and students do not need to bring any card. This method is more effective to prevent problem in process getting attendance manually. This system will integrate with Fingerprint Reader or other Fingerprint Reader that available in the market. It's also have software that some friendly faces so that an overall plan to use this system without much trouble.

The objective of this project is to develop reliable attendance recording system based on biometric attendance that can be used to monitor the presence of students. This system can automatically acquire, store and calculate data and student attendance in the personal computer PC or laptop. Teachers will lead a small handheld with finger scanner and students will be pressing their fingers on it to record attendance. This system has got many advantages. Manual data entry can be avoided no proxy presence made. Caregivers can be notified about non-attendance. The main objective of this project is to monitor the attendance of students in lecture sessions, tutorials and laboratory and others in a more effective way.

II. METHODOLOGY

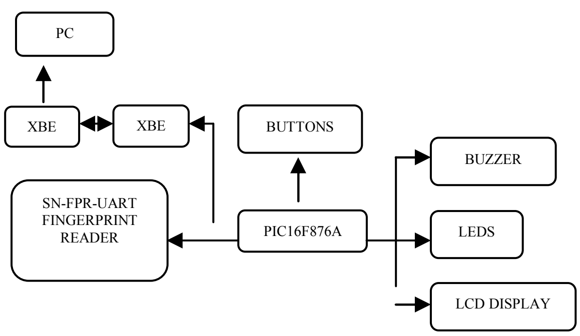

Figure1: Block Diagram Hardware Interface

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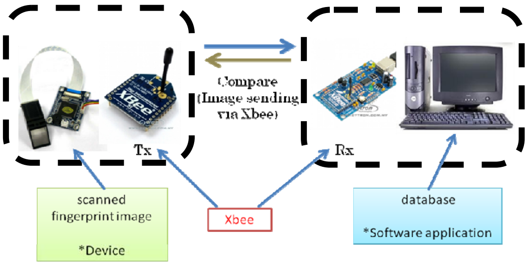

Figure 2: Block Diagram of XBee connection

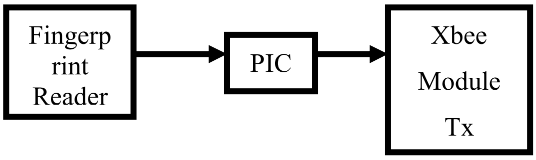

Figure 5: Transmitter block Diagram

This block consists of a finger print reader that records a person's fingerprint data. The optical sensor module forms the core part of the fingerprint. This reader is connected PIC and PIC will convert the data to the XBee module. Then the pattern of finger captured data is sent through the transmitter XBee module for further processing.

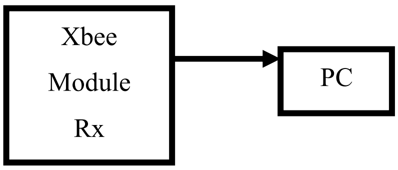

Figure 6: Receiver Block Diagram

Finger pattern captured data received by the XBee receiver module and submitted to the database for someone who has been stored in the system database. If they match then the data will be displayed on the PC through a very user friendly GUI. All data received will be stored in the database.

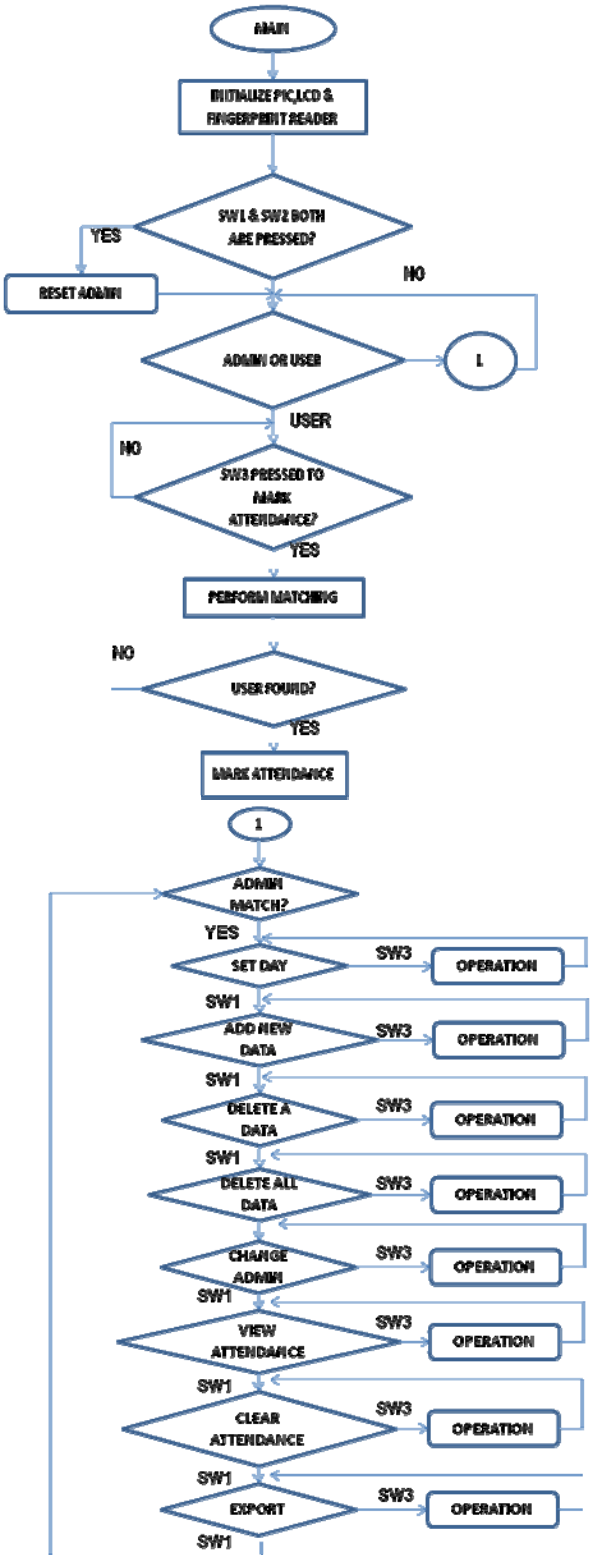

Figure 7: Flow Chart of The System

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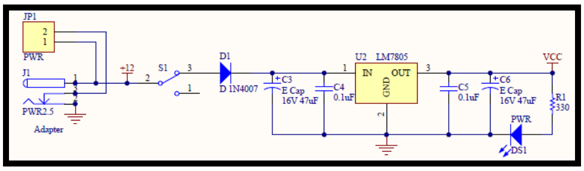

Figure 8: Power Supply

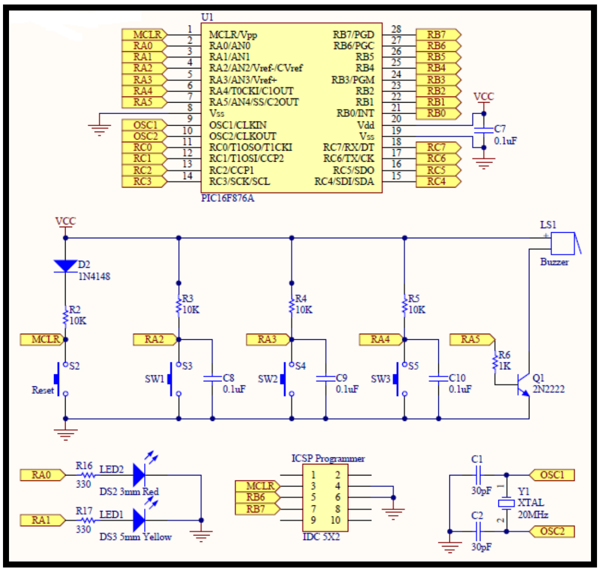

Figure 9: PIC16F876A Connection With Buttons, LEDs and Buzzer

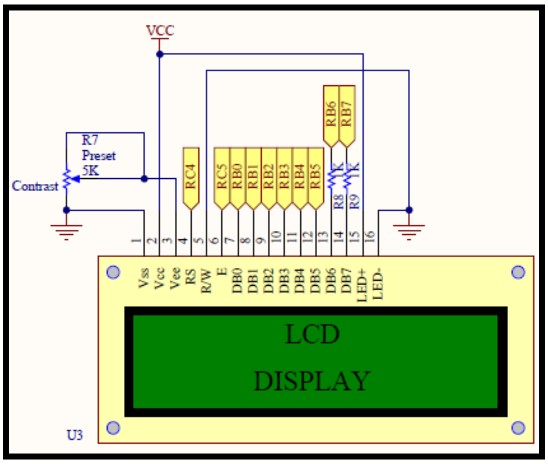

Figure 10: Interface LCD (2 x 16 Characters) with PIC16F876A

Pin	Name	Pin function	Connection
1	VSS	Ground	GND
2	VCC	Positive supply for LCD	5V
3	VEE	Contrast adjust	Connected to a preset to adjust contrast
4	RS	Select register, select instruction or data register	RC4
5	R/W	Select read or write	GND
6	E	Start data read or write	RC5
7	DB0	Data bus pin	RB0
8	DB1	Data bus pin	RB1
9	DB2	Data bus pin	RB2
10	DB3	Data bus pin	RB3
11	DB4	Data bus pin	RB4
12	DB5	Data bus pin	RB5
13	DB6	Data bus pin	RB6
14	DB7	Data bus pin	RB7
15	LED+	Backlight positive input	5V
16	LED-	Backlight negative input	GND

Table.1: LCD Connection Pins and Function Of Each Pin

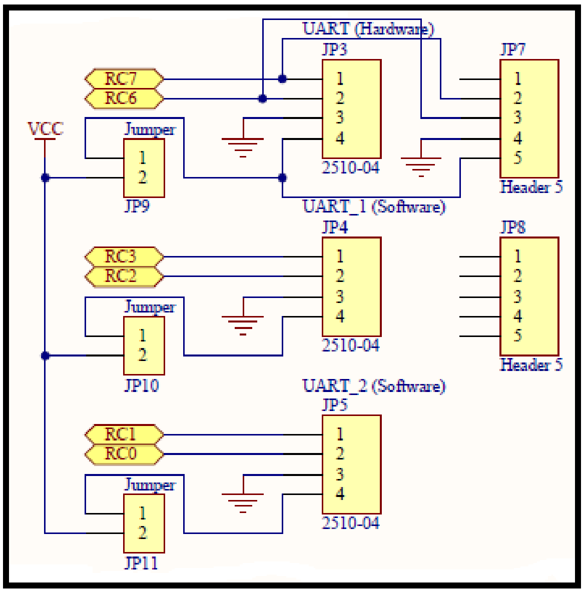

Figure 11: Ports for UART Interface

III. Results and Discussion

1. MPLAB Output Simulation

Figure 12: The overall view of the hardware

When the circuit is switch on the LCD will display the menu which are include of two modes. The first is admin mode and the second is user mode. In the admin mode have eight selections and all the selection only control by the admin. So the user mode cannot go through the selection without the admin permission.

a. Admin Mode

Admin are required to press SW1 or SW2 to SELECT desired operation. Press on SW3 to CONFIRM the selection. Then press on RESET to change to "User Mode"

Figure 13: Sample Mplab Output (MENU)

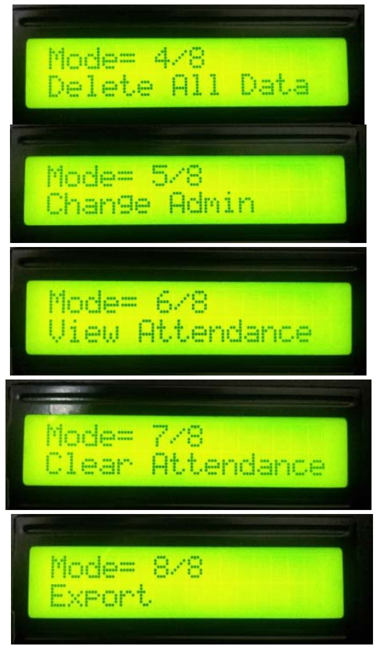

Figure 14: Sample Mplab Output (SELECTION from ADMIN MENU)

b. User Mode

LCD shows "Today is Day # Press 3 to mark" where "#" represents a numeric value from 1-5. If "/" showed in the sentence, it means Admin didn't set the day properly. Registered users are needed to press SW3 and scan their fingers to mark their attendance on the day showed. If attendance marked successfully, LCD will then show "Hello User # Welcome" where "#" represents a numeric value for user ID. Yellow LED will be on for few seconds. Unsuccessful case, LCD will show "Access Denied" and Red LED ON for few seconds also the Buzzer ON for few seconds. Then press RESET to change to "Admin mode".

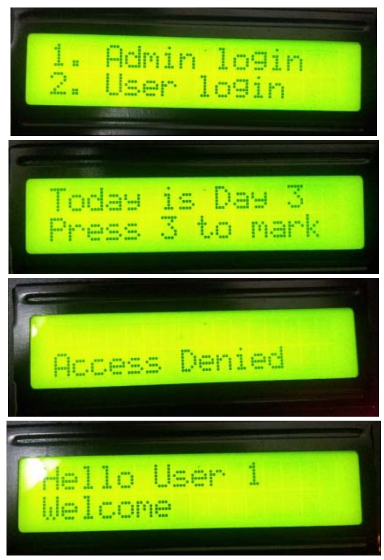

Figure 15: Sample Mplab Output (USER MENU)

2. GUI

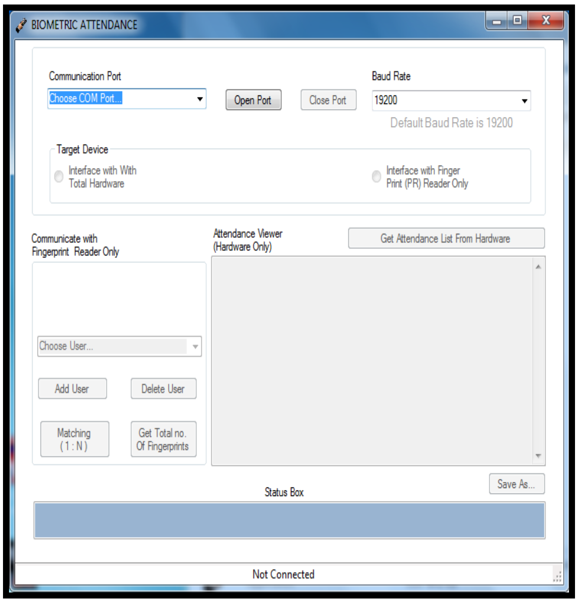

Figure 16: Attendance View before Export to GUI

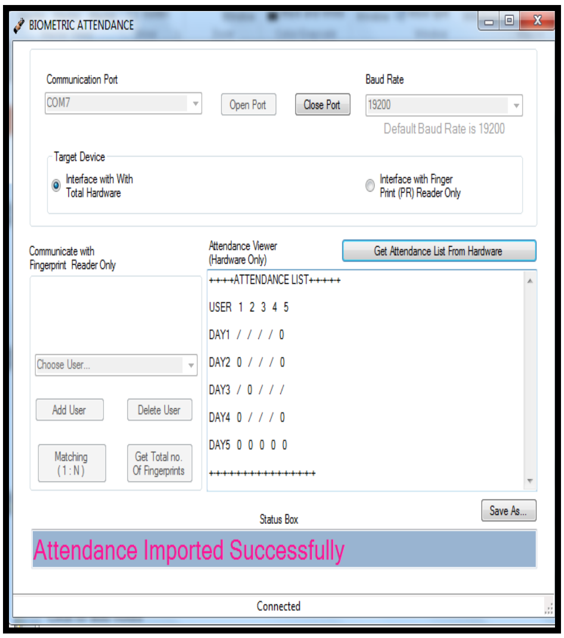

Figure 17: Attendance View before Export to GUI

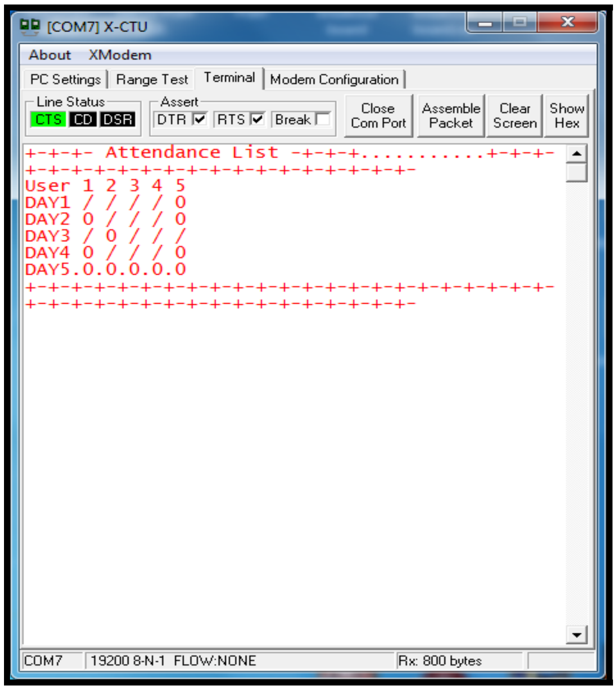

Figure 18: Attendance list in HyperTerminal

IV. CONCLUSIONS AND RECOMMENDATIONS

Biometric technology is an effective tool to verify identity and detect fraud. Matching performance was found to be high. The result based on two fingerprints can be identified well in identifying the user. Analysis confirmed that the biometric data can be set and confirm the identity of the applicant. Expanding the use of biometrics will enhance the ability to detect fraud in the presence of students in the class.

In terms of performance and efficiency, this project has provided a simple sign of the method compared with traditional methods attendance system. By using the database, the data is more structured. This system is also a user-friendly system as data manipulation and retrieval can be done through the interface, making the system universal presence. Therefore, it can be implemented either in academic institutions or in organizations.

Finally, the presence of these systems can be improved by adding features which indicates the presence of the system when an employee or a student late for work or class is possible.

a) The future enhancements that can be developed from this project are to extend the current database to store the complete bio-data of the student.

b) Create a simple system modified, if there is a new student it can facilitate lecturers to complete data entry students.

c) T he system can be made to track the incoming and outgoing time of the students whole day thus clocking in and out helps in additional monitoring.

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